Exercise 1

SQL Fundamentals.

Selecting and Sorting.

01

```
SELECT *
FROM employees.
```

Output:

Employees

ID	First name	Last name	Department	Salary	Hire_date	City
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Bnan	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	Mokoena	Marketing	50 000	2021-11-05	Pretona
5	Eva.	Naidoo	IT	72 000	2017-09-30	Johannesburg

02 To find all unique departments.

```
SELECT distinct department
FROM employees;
```

Department
IT
HR
Finance
Marketing

03 To retrieve first and last names ordered by descending.

```
SELECT first-name,
       Last_name
FROM employees
ORDER BY Salary Desc;
```

First_name	Last_name
Eva	Naidoo
Alice	Green
Cathy	Zuly
David	Mokoena
Brian.	Lee.

04 To retrieve the top 3 highest-paid employees.

```
SELECT ID,
        first_name,
        last_name,
        Salary
FROM Employees
ORDER BY salary Desc
LIMIT 3;
```

ID	first_name	last_name	Salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy.	Zulu.	65 000.

Filtering with WHERE

05 To find employees in the IT department

```
SELECT ID,
        first_name,
        last_name,
        department,
FROM employees,
WHERE department = 'IT';
```

ID	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

06 To find employees in Finance with salary >60 000.

```
SELECT ID,
       first_name,
       last_name,
       department,
FROM employees
WHERE department = 'finance' AND salary >60 000;
```

ID	first_name	last_name	department	Salary
3	Cathy	Zulu	Finance	65 000

07 To find employees in HR or Marketing

```
SELECT ID,
       first_name,
       last_name,
       department,
FROM employees
WHERE department = 'HR' or 'Marketing';
```

ID	first_name	last_name	department
2	Bnan	Lee	HR
4	David	Mokoena	Marketing

08 To find employees not in IT

```
SELECT ID,
       first_name,
       last_name,
       department,
FROM employees
WHERE department NOT 'IT';
```

ID	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David.	Mokoena.	Marketing

09. To find employees in IT, HR, or Finance using IN

```
SELECT ID,
       first_name,
       last_name,
       department,
FROM employees
WHERE department IN ('IT', 'HR') or IN ('Finance').
```

ID	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva.	Naidoo	IT

10. To find employees in IT with salary > 65000 and city Johannesburg.

```
SELECT ID,
       first_name,
       last_name,
       department,
       Salary,
       City,
FROM employees
WHERE department = 'IT' AND Salary > 65000 AND City = 'Johannesburg'
```

ID	first_name	last_name	department	Salary	City
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naidoo	IT	72 000	Johannesburg